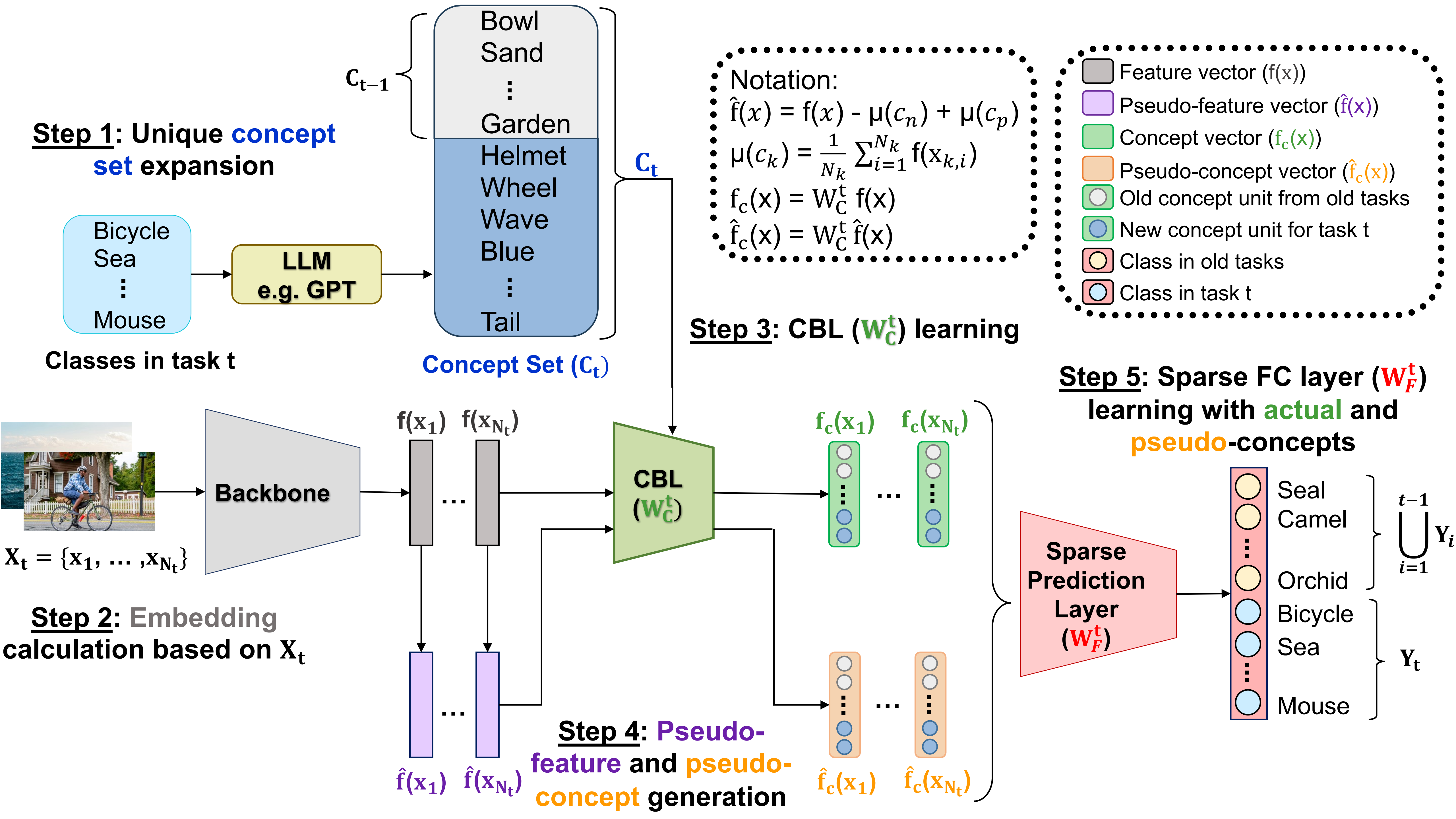

Step 1: Unique concept set expansion
Bicycle
Sea
Mouse
Classes in task t
LLM
e.g. GPT
Bowl
Sand
Garden
Helmet
Wheel
Wave
Blue
Tail
C_{t-1}
C_t
Concept Set (C_t)
Notation:
$\hat{f}(x) = f(x) - \mu(c_n) + \mu(c_p)$
$\mu(c_k) = \frac{1}{N_k} \sum_{i=1}^{N_k} f(x_{k,i})$
$f_c(x) = W_C^t f(x)$
$\hat{f}_c(x) = W_C^t \hat{f}(x)$
Feature vector (f(x))
Pseudo-feature vector ($\hat{f}(x)$)
Concept vector ($f_c(x)$)
Pseudo-concept vector ($\hat{f}_c(x)$)
Old concept unit from old tasks
New concept unit for task t
Class in old tasks
Class in task t
Step 3: CBL (W_C^t) learning
Step 5: Sparse FC layer (W_F^t) learning with actual and pseudo-concepts
$X_t = \{x_1, \ldots, x_{N_t}\}$
Backbone
$f(x_1)$
$f(x_{N_t})$
CBL (W_C^t)
$f_c(x_1)$
$f_c(x_{N_t})$
Sparse Prediction Layer (W_F^t)
Seal
Camel
Orchid
Bicycle
Sea
Mouse
$\bigcup_{i=1}^{t-1} Y_i$
Y_t
Step 2: Embedding calculation based on X_t
$\hat{f}(x_1)$
$\hat{f}(x_{N_t})$
Step 4: Pseudo-feature and pseudo-concept generation
$\hat{f}_c(x_1)$
$\hat{f}_c(x_{N_t})$